

**SAFETY DATA SHEET**

Revision Date 29-Mar-2024

Revision Number 3

**1. Identification**

**Product Name** 2,3-Dichloro-1-propene

**Cat No. :** L08943

**CAS No** 78-88-6  
**Synonyms** 2,3-Dichloropropene; 2,3-Dichloropropylene.

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

**Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No. **US:**001-800-424-9300 / **Europe:**001-703-527-3887

**2. Hazard(s) identification****Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |             |
|--------------------------------------------------|-------------|
| Flammable liquids                                | Category 2  |
| Acute oral toxicity                              | Category 4  |
| Acute dermal toxicity                            | Category 4  |
| Acute Inhalation Toxicity - Vapors               | Category 4  |
| Skin Corrosion/Irritation                        | Category 2  |
| Serious Eye Damage/Eye Irritation                | Category 1  |
| Germ Cell Mutagenicity                           | Category 1B |
| Specific target organ toxicity (single exposure) | Category 3  |
| Target Organs - Respiratory system.              |             |

**Label Elements**

**Signal Word**  
Danger

**Hazard Statements**

Highly flammable liquid and vapor  
Causes skin irritation  
Causes serious eye damage  
May cause respiratory irritation  
May cause genetic defects  
Harmful if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Keep cool

**Response**

IF exposed or concerned: Get medical attention/advice

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

Call a POISON CENTER or doctor/physician if you feel unwell

If skin irritation occurs: Get medical advice/attention

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

Wash contaminated clothing before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER or doctor/physician

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell

Rinse mouth

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store locked up

Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Harmful to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component           | CAS No  | Weight % |
|---------------------|---------|----------|
| 2,3-Dichloropropene | 78-88-6 | 98       |

### 4. First-aid measures

|                                            |                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                    |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention.                                                         |
| <b>Inhalation</b>                          | Remove from exposure, lie down. Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Immediate medical attention is required. |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water. Call a physician immediately. If possible drink milk afterwards.               |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Causes eye burns. . Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting                     |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                |

### 5. Fire-fighting measures

|                                         |                                                                                                                                                                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Alcohol resistant foam. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                                                                                        |
| <b>Flash Point</b>                      | 10 °C / 50 °F                                                                                                                                                                                                                                   |
| <b>Method -</b>                         | No information available                                                                                                                                                                                                                        |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                                                                                                        |
| <b>Explosion Limits</b>                 |                                                                                                                                                                                                                                                 |
| <b>Upper</b>                            | 7.8%                                                                                                                                                                                                                                            |
| <b>Lower</b>                            | 2.6%                                                                                                                                                                                                                                            |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                                                                                        |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                                                                                        |

#### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### Hazardous Combustion Products

Thermal decomposition can lead to release of irritating gases and vapors. Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
2

**Flammability**  
3

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Remove all sources of ignition. Take precautionary measures against static discharges.                                                                                                                                                                                                                   |
| <b>Environmental Precautions</b>            | See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage. Do not flush into surface water or sanitary sewer system.                                                                                                                                      |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Flammables area. Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Strong oxidizing agents. Strong bases. Oxidizing agent.                                                                                                                                                   |

## 8. Exposure controls / personal protection

|                                             |                                                                                                                                                                                                                                       |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Exposure Guidelines</u></b>           | This product does not contain any hazardous materials with occupational exposure limitsestablished by the region specific regulatory bodies.                                                                                          |
| <b>Engineering Measures</b>                 | Ensure adequate ventilation, especially in confined areas. Ventilation systems. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. |
| <b><u>Personal Protective Equipment</u></b> |                                                                                                                                                                                                                                       |
| <b>Eye/face Protection</b>                  | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.                                                           |
| <b>Skin and body protection</b>             | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                             |
| <b>Respiratory Protection</b>               | Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions.                                                                              |
| <b>Recommended Filter type:</b>             | Organic gases and vapours filter. Type A. Brown. conforming to EN14387.                                                                                                                                                               |
| <b>Hygiene Measures</b>                     | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                |

## 9. Physical and chemical properties

|                            |                             |
|----------------------------|-----------------------------|
| <b>Physical State</b>      | Liquid                      |
| <b>Appearance</b>          | Colorless                   |
| <b>Odor</b>                | sweet                       |
| <b>Odor Threshold</b>      | No information available    |
| <b>pH</b>                  | No information available    |
| <b>Melting Point/Range</b> | 10 °C / 50 °F               |
| <b>Boiling Point/Range</b> | 94 °C / 201.2 °F @ 760 mmHg |
| <b>Flash Point</b>         | 10 °C / 50 °F               |
| <b>Evaporation Rate</b>    | No information available    |

|                                        |                          |
|----------------------------------------|--------------------------|
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | 7.8%                     |
| Lower                                  | 2.6%                     |
| Vapor Pressure                         | 61.2 mmHg @ 25 °C        |
| Vapor Density                          | 3.8 (Air = 1.0)          |
| Specific Gravity                       | 1.200                    |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| Molecular Formula                      | C3 H4 Cl2                |
| Molecular Weight                       | 110.97                   |

## 10. Stability and reactivity

|                                  |                                                                                                                                                          |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                                                               |
| Stability                        | Stable under normal conditions.                                                                                                                          |
| Conditions to Avoid              | Incompatible products. Keep away from open flames, hot surfaces and sources of ignition.                                                                 |
| Incompatible Materials           | Strong oxidizing agents, Strong bases, Oxidizing agent                                                                                                   |
| Hazardous Decomposition Products | Thermal decomposition can lead to release of irritating gases and vapors, Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen chloride gas |
| Hazardous Polymerization         | No information available.                                                                                                                                |
| Hazardous Reactions              | None under normal processing.                                                                                                                            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component           | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation |
|---------------------|--------------------------|------------------------------|-----------------|
| 2,3-Dichloropropene | LD50 = 320 mg/kg ( Rat ) | LD50 = 1580 mg/kg ( Rabbit ) | Not listed      |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component           | CAS No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------|---------|------------|------------|------------|------------|------------|
| 2,3-Dichloropropene | 78-88-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

|                                                   |                                                                                                                     |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>STOT - single exposure</b>                     | Respiratory system                                                                                                  |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                          |
| <b>Aspiration hazard</b>                          | No information available                                                                                            |
| <b>Symptoms / effects, both acute and delayed</b> | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                            |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information.  |

## 12. Ecological information

### Ecotoxicity

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

|                                      |                           |
|--------------------------------------|---------------------------|
| <b>Persistence and Degradability</b> | No information available  |
| <b>Bioaccumulation/ Accumulation</b> | No information available. |
| <b>Mobility</b>                      | No information available. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                      |        |
|----------------------|--------|
| <b>UN-No</b>         | UN2047 |
| <b>Hazard Class</b>  | 3      |
| <b>Packing Group</b> | II     |

### TDG

|                      |        |
|----------------------|--------|
| <b>UN-No</b>         | UN2047 |
| <b>Hazard Class</b>  | 3      |
| <b>Packing Group</b> | II     |

### IATA

|                             |                  |
|-----------------------------|------------------|
| <b>UN-No</b>                | UN2047           |
| <b>Proper Shipping Name</b> | DICHLOROPROPENES |
| <b>Hazard Class</b>         | 3                |
| <b>Packing Group</b>        | II               |

### IMDG/IMO

|                             |                  |
|-----------------------------|------------------|
| <b>UN-No</b>                | UN2047           |
| <b>Proper Shipping Name</b> | DICHLOROPROPENES |
| <b>Hazard Class</b>         | 3                |
| <b>Packing Group</b>        | II               |

## 15. Regulatory information

### United States of America Inventory

| Component           | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|---------------------|---------|------|-----------------------------------------------|-----------------------------|
| 2,3-Dichloropropene | 78-88-6 | X    | ACTIVE                                        | -                           |

### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed  
 '-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b) - Notices of Export**

Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component           | CAS No  | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|---------------------|---------|-----|------|-----------|-------|------|------|------|-------|------|
| 2,3-Dichloropropene | 78-88-6 | X   | -    | 201-153-8 | -     | -    | X    | -    | X     | -    |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations**
**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component           | CAS No  | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|---------------------|---------|----------|-------------------------------|---------------------------------|
| 2,3-Dichloropropene | 78-88-6 | 98       | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component           | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|---------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| 2,3-Dichloropropene | X                          | -                           | -                      | -                         |

**Clean Air Act**

Not applicable

OSHA - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component           | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|---------------------|--------------------------|-------------------------------------------|-------------------------------|
| 2,3-Dichloropropene | 100 lb                   | -                                         | 100 lb<br>45.4 kg             |

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component           | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------------|---------------|------------|--------------|----------|--------------|
| 2,3-Dichloropropene | X             | X          | X            | -        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
 DOT Marine Pollutant N  
 DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH**

| Component           | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 2,3-Dichloropropene | 78-88-6 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component           | CAS No  | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------|---------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2,3-Dichloropropene | 78-88-6 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component           | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2,3-Dichloropropene | 78-88-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y45                      |

**16. Other information****Prepared By**

Health, Safety and Environmental Department  
 Email: chem.techinfo@thermofisher.com  
 www.thermofisher.com

**Revision Date**

29-Mar-2024

**Print Date**

29-Mar-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**